current is a measure of absorbed dose. The effect is referred to as RITAC (radiation induced thermally activated current) and the technique is named TC (thermocurrent) dosimetry. Because it is instrumentally easier to measure an electric current than light, the TC dosimeter may replace the TLD with time.

Doses can also be calculated from the product of the dose rate and the time of exposure. The most common *dose rate meter* is the ionization chamber. Because of the close connection between this instrument and pulse type ionization counters, which measure individual nuclear particles entering the detector, the discussion of ionization chambers is deferred to Ch. 9, which deals in more detail with radiation measurement techniques.

If the number, energy, and type of nuclear particles being absorbed in a material can be measured or estimated, the absorbed dose can be calculated as described previously. Such calculations are very important, particularly in the field of radiation protection.

8.11. LARGE-SCALE NON-BIOLOGICAL APPLICATIONS

In this section we deal only with the more important non-biological applications, as the latter are treated in Ch. 15, which specifically deals with biological effects of radiation.

Ionizing radiation produces ionized and excited atoms and molecules in all materials. Excited molecules formed directly or by recombination reactions between electrons and cations decompose in the vast majority of systems to highly reactive free radicals. The reactive species formed on radiolysis are precursors of further reactions, such as reduction, oxidation, polymerization, cross linking and so on. It should therefore be possible to apply radiation chemical methods to industrial processes and, consequently, extensive applied research and development of radiation chemistry has been carried out during the past three decades.

A large proportion of the radiation induced reactions can be brought about by thermal, photochemical or chemical initiation, but the advantages of radiation initiation are generally claimed to be:

- No contamination by catalyst and catalyst residue.
- Temperature independence.
- Easy control of radiation intensity and hence rate of induced reactions.
- High speed treatment capability.
- Ionizing radiation offers the advantage of greater penetrating power compared to initiation by UV-light.

8.11.1. Radiation sources

The radiation sources normally used can be divided into two groups, those employing radioactive isotopes (e.g. ^{60}Co) and those employing electron beam accelerators (EBA).

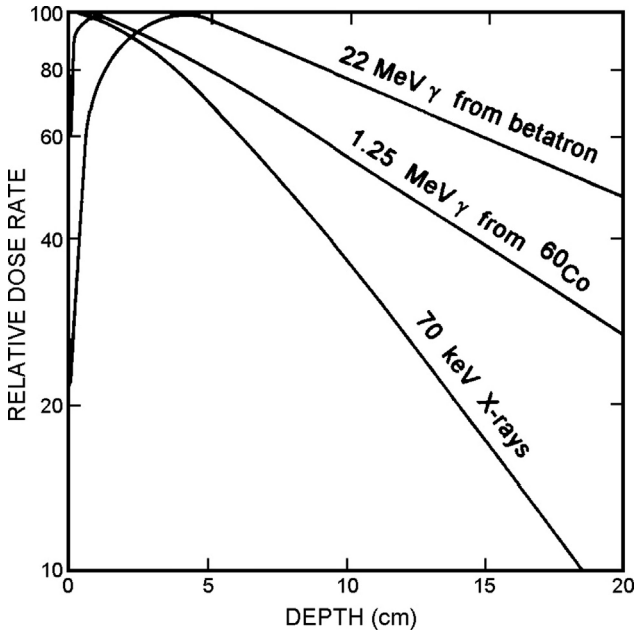


Figure 8.9 Depth-dose curves for electromagnetic radiation in water. Distance between source and water surface is 0.8 m; area of beam at water surface 0.01 m^2. *(From Spinks and Woods.)*

Gamma radiation, having a large penetration range (Fig. 8.9), is advantageously used for initiation of reactions in solids and liquids, food irradiation and sterilization of medical products. The useful penetration depth of electrons is approximately given by $0.4.E/\rho$ (cm), where E is the electron energy (in MeV) and ρ the density of the irradiated material (in g/cm^3). For EBA radiation, dose rate and penetration depth are easily controlled by varying the beam current and acceleration voltage. Very high radiation intensity can be provided and hence be applied to high speed processes. Radiation absorption is discussed in detail in Ch. 7, and particle accelerators in Ch. 16

8.11.2. Process criteria

The radiation dose required to completely convert 1 kg material to product is

$$D = (M\ G)^{-1}\ \mathrm{J\ kg^{-1}} \tag{8.41}$$

where D is the radiation dose (Gy), G is the radiation yield ($\mathrm{mol\ J^{-1}}$), and M is the molar mass ($\mathrm{kg\ mol^{-1}}$).

Ionizing radiation is an expensive form of energy, whether the radiation source is ^{60}Co or an accelerator, and generally at least one of the following criteria are fulfilled in the established radiation processes:

- A small amount of chemical change causes a marked change in physical properties (e.g. for polymers or biological systems).

- The radiation induced reaction has a high yield (e.g. polymerization, chain reaction).
- The radiation has a specific effect or process technical advantages which not easily can be obtained by other methods (e.g. staining of glass).

About 80% of ongoing radiation processes can be ascribed to the first category. Some examples of radiation processing are briefly described in the remainder of this chapter.

8.11.3. Radiation induced synthesis

A great number of reaction types have been investigated, the majority involve radical initiated organic chain reactions. A limited number of radiation induced synthesis has, however, been developed to the pilot plant or industrial scale. Examples of reported industrial synthesis are sulfoxidation and sulfochlorination of hydrocarbons for detergent production and polymerization of ethene.

The bromoethane process, which from 1963 and a number of years onwards, was used by the Dow Chemical Company to produce 500 tons bromoethane per year is of especial interest. In the process a 67 GBq ^{60}Co-source was utilized to irradiate hydrogen bromide-ethane mixtures and initiate the reaction sequence

Initiation

$$HBr \sim\sim\sim\sim\rightarrow H\cdot + Br\cdot \quad (8.42)$$

$$C_2H_4 \sim\sim\sim\sim\rightarrow C_2H_2 + 2\,H\cdot \quad (8.43)$$

$$H\cdot + HBr \rightarrow H_2 + Br\cdot \quad (8.44)$$

Propagation

$$Br\cdot + C_2H_4 \rightarrow \cdot CH_2CH_2Br \quad (8.45)$$

$$\cdot CH_2CH_2Br + HBr \rightarrow CH_3CH_2Br + Br\cdot \quad (8.46)$$

Termination

$$2\,Br\cdot \rightarrow Br_2 \quad (8.47)$$

$$2\cdot CH_2CH_2Br \rightarrow BrCH_2CH_2CH_2CH_2Br \quad (8.48)$$

The chemical yield was, depending on reaction conditions, in the range 0.001–0.01 mol/J.

8.11.4. Industrial radiation processing

Today a large number of ^{60}Co and ^{137}Cs based irradiation sources and electron beam accelerators (EBA) are in use. In the following some of the more important uses will be summarized.

Sterilization: There has been a steadily increasing demand for single use disposable medical supplies (e.g. injection syringes, injection needles, surgical gloves) as well as surgical implants, contact lens solution and laboratory animal feed. Radiation

sterilization doses in the range 25–50 kGy have been adapted in many countries. Both ^{60}Co and high energy electron accelerators are used as radiation sources. Radiation sterilization has the advantage of avoiding the use of toxic chemicals (e.g. ethylene oxide) and high temperature. A further advantage is that the material can be sterilized in a continuous process after final packaging.

Wire and cable: Thermoplastics and elastomers are widely used as electrical insulating material due to their physical properties and processability. Cross linking is an effective means for improving e.g. the thermal resistance and tensile strength. EBA-irradiation ($\approx$50 kGy) affords a rapid, well controlled cross linking and is used by several major producers of thin wires and cables.

Shrinkable film and tubing: Cross linked semi-crystalline thermoplastics display rubberlike properties at temperatures above their melting points. On deformation followed by fast cooling the polymer maintains its deformed shape. The polymer returns to its original shape when reheated. This memory effect is applied in the production of heat shrinkable films and tubing. Radiation doses of the order of 40–100 kGy are used in the production of heat shrinkable products.

Curing of surface coatings and inks: Irradiation is used to cure printing inks and varnishes and to bond coatings to surfaces of paper, fabrics, polymer films and steel. The absorbed doses are in the range 10–100 kGy and use of accelerators with low energy and high beam current affords selective energy deposition in the surface layer and high product throughput. Industrial radiation curing lines are operating in the car, wood panel and steel industries. Due to the low electron energy, no heavy shielding is required and the accelerators can easily be placed in normal industrial environment.

Potential applications: Immobilization or trapping of bioactive material such as enzymes, antigens, antibodies and drugs in polymer matrixes by radiation processes have received much attention.

Radiation treatment: To remove organics and metallic pollutants from waste water and SO_x and NO_x from flue gases emitted from coal power stations and industrial plants have been studied in great detail. The cleaning of flue gases has been developed to the pilot plant stage.

8.12. TECHNICAL USES OF SMALL DOSE-RATES

The ability of nuclear radiation to ionize gases is used in several applications. Static electricity can be eliminated through the installation of an α- or β-source in microbalances. Similarly, but on a much larger scale, elimination of static electricity is used in the paper, textile and paint industries. An ionization instrument for the analysis of gas has been developed in which the gas passes through a small chamber where it is irradiated by a small radioactive source. For a constant source of radiation, the ions produced in the gas depend on the flow velocity of the gas and on its temperature, pressure and atomic